NEW ALEVEL SCENARIO BASED ITEMS

P425/2

UACE PRINCIPAL MATHEMATICS MOCKS

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THEME 1: NUMERICAL ANALYSIS

ITEM 1

A construction company has been awarded a contract to install a drainage channel alongside a newly constructed straight road. For planning purposes, the engineers represent the drainage channel by the curve $y = x^3 - 7x + 6$, while the road is represented by the x-axis, $y = 0$.To determine the points where the drainage channel intersects the road, the engineers intend to sketch the graph of the curve and estimate the three points of intersection within the interval $-4 \leq x \leq 3$.

In addition, the company plans to construct public washrooms in the region enclosed by the drainage channel and the road between $x = 2$ and $x = 3$. To assess whether the available land is sufficient, the engineers will estimate the enclosed area with six ordinates. If the estimated area is less than 4 square units, the site will be considered too small for the proposed washrooms.

The company also intends to purchase cylindrical concrete culverts for the drainage system. The quoted dimensions of each culvert are; radius of 80.45 cm and a height of 120.2 cm correct to the nearest decimal place. Before placing the order, the engineers wish to verify the reliability of these measurements. They will determine the percentage error in the volume of the culvert and proceed with the purchase only if the percentage error is less than 1.0%.

Task.

Help the construction company make informed decisions using the information above.

(UNNASE)

ITEM 2

Mr. Mukasa, a farmer in Mubende, a district 150 km from Kampala intends to start a fish farming project. The government through the Parish Development Model (PDM), provides free fish to qualified farmers.

The P.D.M officials require that for a farmer to qualify, they must have a pond with surface area of at least 110 m^2.

The farmer's only piece of land that he wants to use for this project is irregular in shape, and he intends to find out if it meets the official's requirements.

He contacted his engineer from Kampala who is to travel by public means. The engineer left Kampala at 8:00 am. On his way, he noticed that it took him 1.5 hours to reach Mityana town, which is 80 km from Kampala He reached Kassanda town, which is 120 km from Kampala at 10:30 am.

He is expected to reach Mubende by 12:00 pm before Mr. Mukasa leaves for other duties.

The engineer established the surface area of the land by measuring it's width at intervals of 4 m along its 24 m length. The measurements are as shown below;

$Distance\ from\ one\ end(xm)$	0	4	8	12	16	20	24
$Width\ of\ the\ pond\ (y\ m)$	3.0	4.2	5.1	5.8	5.4	4.6	3.5

When the officials arrived, they accurately measured the area of the piece of land as 115 m2. Mr. Mukasa would consider the engineer's estimation "excellent" if it had a relative error not greater than 0.02.

Task

(a) Establish whether the engineer found Mr. Mukasa on site.
(b) Based on the engineer's estimation, decide whether Mr. Mukasa would qualify for the P.D.M funds.
(c) Was the engineer's work rated "excellent"?

(WAKISSHA)

ITEM 3

The Uganda National Roads Authority (UNRA) is upgrading a 50km road that connects a trading centre to a lake community. A Sino truck transports gravel from a quarry and off loads it at different points along the road according to the condition of the road surface.

The following offloading records were made along the route. At the trading centre 4 tonnes were offloaded. At the swampy area 40km from the community 3.4 tonnes were offloaded. At the beginning of the hill 20km from the centre 2.8 tonnes were offloaded. 30km from the centre 2.2 tonnes were offloaded. At a distance of 10km from the lake community 1.6 tonnes were offloaded and at the lake community 1.0 tonnes were offloaded. The roads supervisor wants to plan the distribution of stones more effectively for the remaining sections of the road. The engineer wanted to offload stones at a distance of 9km from the community and also wanted to offload 2.09 tonnes of stones

Engineers also established the relationship between the quantity of stones offloaded, y tonnes, and the distance x in Kilometers as $e^x cosx + 1 = 0$ for the section $1 \leq x \leq 2$ (kilometers) for the road design in that particular area. The engineer wanted to perform a dry run to identify actual location to a degree of accuracy of two decimal places.

Task

(a) Advise the supervisor on the quantity of stones to be offloaded and the location for offloading from the centre.

(b) (i) Help the supervisor to establish the road design graphically, and distance where offload was made.

ii) Advise the engineer on the most accurate location.

(JJEB)

ITEM 4

An engineer is designing a new town road. The layout of the road follows the path $y = 2x + 7cos2x$. This road must cross a straight river located at $y = 0$. To plan bridge construction, the engineer needs to find where the road and river intersect. The engineer wants to show graphically that the road meets the river at a point between 2km to 3km measured from one boundary B_1 of the town. To pinpoint the exact spot, he plans to use an iterative formula based on the Newton Raphson Method until his answer stabilizes. The difference between two consecutive approximations must be less than a tolerance threshold of 0.0001 km. The town will only approve the project if the determined approximate location of the bridge falls within a safety zone between 2.0440 km and 2.0452 km measured from the boundary B1.

The engineer also plans to first visit the location of the river to make some observations. He plans to use a bus that moves between the boundaries of the town along this road. The bus starts from B1 and makes stops at different stages 500m apart the bus fares up to the first, second, third and fourth stages are shs11,000, shs15,000, shs18,500 and shs20,000 respectively. The engineer has established that he can use interpolation to approximate the fare to the bridge location which 1800m from B_1 . The engineer wishes to spend at most shs42,000 to and from back to B1 or else he postpones the visit for the following day.

The engineer also wants to use an approximate method to determine the area enclosed by the road and the river which is described $\int (2x + 7cos2x)dx$ from $x = 3km$ to $x = 4km$ using 6 ordinates but he has been opposed by some of his colleagues claiming that this will result into an error of more than 1% and this will give wrong conclusions on the costs involved in the contract. The engineer wishes to determine the percentage error in this area to ascertain whether his friends are right so that he can address the fear of his colleagues.

Task:

Help the engineer obtain the information he needs to make informed decisions.

(ACEITEKA)

ITEM 5

In a certain school, the Headteacher wishes to improve security around the corners of the compound by installing security cameras. He contacted a technician to identify suitable positions for the installation. Upon mapping, the technician established an initial position and drew a baseline through the compound. He realized that the path across the compound

is modelled by the function $f(x) = 2cosx - x^2$, where x is the distance in hundreds of meters from the initial position along the baseline.

He also noticed that the path intersects the baseline at two points; the most suitable positions where the cameras are to be installed. However, due to financial constraints, the headteacher prefers the position on the positive side of the initial position. He needs to determine the exact location where the camera should be installed using an iterative method to avoid mis calculation.

Additionally, the headteacher discovered that the area between the path and the baseline can accommodate a tent to serve as a consultation area, provided the area is at least 10,000 square meters. The technician uses a numerical method with 5 sub-intervals along the baseline to ascertain this area.

Task:

(a) Justify the exact position where the headteacher will install his camera.

(b) Is the area of interest being enough for the consultation tent? Make recommendation basing on the technician method.

(UMTA)

ITEM 6

An aerospace engineer is designing a specialized cooling fin for a satellite component. The profile of the fin is modeled by a complex curve and the engineer needs to determine its cross - sectional area which can later help him to determine its weight and heat dissipation. The top boundary of the fin's cross section is defined by the function $f(x) = \frac{10}{x^2+2}$ the width of the fin bases span from $x = 0\ to\ x = 2\ m$ and he hopes to subdivide the interval into four equal sub-intervals. During testing, the same aircraft is used to deliver 20 packages in 45 minutes and another 40 packages in 75 minutes. He is wondering whether to take 60 packages if time does not exceed 105 minutes otherwise, he is to find time before he does the delivery of 60 packages to have the final decision.

The engineer also wishes to derive a formula for estimating the relative error he made in determining the volume of the cylinder of his aeroplane engine, the errors the radius, height and volume being $\Delta r, \Delta h\ and\ \Delta v$ respectively.

Task:

Help the engineer make appropriate decisions using mathematical methods.

(KASSEC)

ITEM 7

A district local government is constructing a community water supply system to provide clean water to a newly established health centre. Before approving the construction, the District Engineer must verify the accuracy of the proposed water storage tank dimensions, assess the adequacy of the drainage channel, and confirm the required diameter of the main supply pipe.

The tank has a radius of 2.5 m, correct to one decimal place. The engineer estimates that the maximum relative absolute error in the calculated volume of the tank is 0.125. Before approving construction, the engineer wishes to determine whether the measured height of the tank is sufficiently accurate. If the maximum relative absolute error in the measured height exceeds 0.10, the height of the tank will be measured again before construction begins. Otherwise, the existing measurements will be approved for construction.

To protect the water tank from flooding during the rainy season, the District Engineer designed a drainage channel. The cross-sectional profile of the channel is modelled by

$$y = \frac{50x + 27\tan^{-1} x}{1 + x^2}$$

where x and y are measured in metres.

The engineer recorded the vertical distances (ordinates) from the ground level to the channel profile at five equally spaced positions between $x = 0\ and\ x = 6$, inclusive, and used these measurements to estimate the cross-sectional area correct to three decimal places. If the estimated cross-sectional area exceeds 120 m^2, the drainage channel cross-section will be enlarged to improve storm-water discharge. Otherwise, the current drainage design will be maintained.

To ensure adequate water pressure throughout the distribution network, the internal diameter (in centimetres) of the main supply pipe must satisfy the equation:

$$e^{lnx} - cosec\ x = 0$$

Preliminary calculations indicate that the required pipe diameter lies between 2.6 cm and 2.9 cm. Before purchasing the pipes, the engineer wishes to obtain a better approximation to the required diameter correct to three decimal places. If the better approximation exceeds 2.8 cm, a high-pressure commercial pipe will be purchased. Otherwise, a standard PVC pipe will be used.

Task:

Using appropriate mathematical methods, advise the District Engineer on the suitability of the proposed designs for the community water supply project before construction begins.

(MATIGO)

ITEM 8

National Water and Sewerage Cooperation (NWSC) is encountered with a problem of the volume capacity of their cylindrical pipes that supply water to their customers. They consulted their engineer who found out that the diameter is 0.62metres and length 6.2metres. There was need to know the range between which the volume of cylindrical pipe lie, the engineer advised the NWSC manager that for the cylindrical pipes to provide sufficient water supply to its customers, the range should be between 1.6 and 2.0.

The NWSC was also encountered with a problem of location of a point at which to locate a pumping machine along a linear bank of a river that stretches from 3km to 5km. Previously data collected by the engineering department had indicated that this area was bounded by a line($y = 4x + 6$) and another curve ($y = 3e^{0.5x}$). The surveying department noticed that using a graphical method the point of intersection between the above two curves that lie along the line from 3km to 5km above would be a suitable approximate point for the location of the pumping machine and required that it should be rounded off to 1 decimal place, use a suitable iterative method to get the exact value, if this point lies between 3.5 and 4, it would be a suitable approximate point at which to install the pumping machine.

The NWSC was also encountered with a problem of establishing the costs and distances involved in transportation of their water installation materials. It had been noted for distances of 30km, 40km, and 50km the costs were sh. 60,000/=, sh. 100,000/= and sh. 130,000/= respectively. They were approached by two of their customers, one paid sh. 150,000/= and another one was to be left at a distance of 46km. There was need to use a suitable linear method to estimate for them the distance and the cost. The second customer complained that he had been overcharged by paying sh. 120,000/= and there was need to justify his claim.

TASK:

As a mathematics student, analyze the problems of NWSC by giving an interpretation, methodology and justification.

(ASSHU-ANKOLE)

ITEM 9

In the process of construction of a water reservoir, a team of surveyors measured the depth profile of land at equal intervals of 10 meters across the proposed site and obtained the following systematic measurements in meters: 2.4, 3.1, 3.8, 4.2, 4.0, 3.5 *and* 2.8. Engineers want to estimate the cross-sectional area of the land before excavation begins.

The engineer further explains that the amount of water expected to be pumped into the reservoir depends on solving the equation $x^3 - 4x - 10 = 0$, where x represents the operating efficiency factor of the pump. A preliminary estimate suggests that the value of x lies between 2 and 3. Since no exact algebraic solution is readily available. Before final construction begins, the supervisor had to measure the planned rectangular concrete base of the reservoir and record the length as $30m \pm 0.1m$ and width as $30m \pm 0.1m$. The project committee was worried that inaccurate measurements may affect the construction budget and mount of materials required.

Tasks;

As a mathematics student, if given a chance to analytically advise on the above situation, how would you come up with;

a) The appropriate numerical method to find the approximate area of the land available for excavation.
b) A suitable iterative method to obtain the solution to 3 decimal places.
c) The range within which the exact area lies and the percentage error in approximating the area

(COUHEIA)

ITEM 10

A certain distribution company uses trucks to carry its luggage from Kampala to Busia. The company introduced a module to monitor the charges in relation to weights, speed and distance covered and fuel consumption by the vehicle. The cost of fuel (y in shillings) for distances (x in kilometers) is as shown in the table below.

$Distance\ x\ (km)$	10	20	30	40
$Cost\ y(shs)$	2800	3600	4400	5200

The manager will hire another driver if the cost of covering 45km exceeds shs. 6000 but if the driver spent shs. 4000, on fuel then the minimum distance to be covered should be at most 30km or else he/she will lose the job.

The route taken by the truck is described by the function $f(x) = x^3 - 5x + 1$, and at every check point, $f(x) = 0$. Unfortunately, the company did not insert this module into their system, yet the only tool left is graph and the driver is confused where exactly the vehicle will stop for police checkpoint, and failure to stop at the right spot, the vehicle will be impounded. The driver is required to be careful only in the first 4 kilometers of the journey since it contains the deadliest checkpoints. His friend is a police man who works at one of the checkpoints between 2km and 3km from the company premises and told the driver to consult their technicians at the main station to iterate the graphical value until appropriate numerical distance is obtained, if not the company should be ready to suffer the loss.

Task.

As a student of mathematics;

(a) Guide the manager on the values he will use in order to hire or maintain the driver.
(b) Help the company on how they can dodge the loss.

(UTES)

ITEM 11

Kato runs a delivery service business around Kampala city centre. The total cost (c) of delivering a package of weight (w) to a distance (d) is modelled by the formula $c = bw + \frac{r}{d}$, where b is the charge for the package delivered per kilometer and r is the distance adjustment rate.
For one customer, Kato is asked to deliver a package of items from the city centre to a certain destination. The weight of the package and the distance to the destination was estimated as 18.5kg and 24km with possible errors of 0.3kg and 1.5km respectively.

Kato charges UGX.2,500 with a margin of error 4% and uses a distance adjustment rate of 2 which tends to vary by 5% depending on the road conditions.

In order to inform the client of how much is needed to deliver the package, Kato needs to know the range with in which the total cost to be incurred by this client is expected to lie.

A certain regular client has used Kato's delivery services on three past occasions. He was charged UGX.32,000 for a distance of 10km, UGX.45,000 for a distance of 16km and UGX.38,000 for a distance of 12km from the city centre. The client now wants Kato to deliver a package to a place 17.5km from the city centre and he has prepared UGX.47,500 for this delivery and he wants to know whether the money will be enough or not.

Basing on his previous experience and expertise in the delivery business, Kato found out that the delivery condition score y for a package can be obtained using the two models $y = 2x$ and $y = 3\cos\frac{1}{2}x$, where x represents the expected delivery time in hours.

The time at which the two models give exactly the same delivery condition score between 1 and 2 is used as the initial estimate of the expected delivery time for a package and using a suitable iterative formula, the initial estimate can be improved until the maximum possible error in the estimate is 0.0005. Kato promised a certain customer that the package will be delivered within 1.5 hours and he must decide whether to start on the journey immediately or first notify the customer that delivery will delay.

Task.
As a mathematician, help Kato find the required range, determine whether the money prepared by the regular client will be enough to cover the total cost for delivering the package and decide whether to start the journey immediately or notify the customer that the delivery will delay.

(MEC)

ITEM 12

At a regional hospital, a team is monitoring the fuel level in a backup generator tank to ensure there is enough fuel during long power outages. The fuel height h (in meters) is measured using a sensor with a possible error of $\pm\ 0.05\ m$. The relationship between fuel height and fuel volume V (in litres) is given by;$V = h^3 + 6h^2 + 10h + 5$. The team recorded the following measurements;

$Fuel\ volume(l)$	22	86	230
$Fuel\ height\ (m)$	1	3	5

During inspection, condensation inside the tank makes the gauge reading at 2m unclear, so the team must estimate the corresponding fuel volume. At the same time, due to an expected prolonged power outage, more fuel may be added beyond the measured range, requiring prediction of the fuel volume at higher fuel levels.

Since the sensor readings are not perfectly accurate, the team must also analyze the possible error in calculated fuel volume.

Finally, to avoid generator shutdown, the team needs to determine the exact fuel height at which the tank reaches the minimum safe fuel volume of 100 litres.

Tasks:

Help the team to;

a) Determine the absolute error in fuel volume when the fuel height is 3m, given that the sensor has a possible error of $\pm 0.05m$.
b) Estimate the fuel volume when the fuel height is 2m and also predict the fuel volume when the fuel height reaches 6m.
c) Use the Newton-Raphson Method to determine the fuel height at which the fuel volume becomes exactly 100 litres correct to 3 decimal places taking initial approximate height $h_o = 3.0$

(TAEB)

ITEM 13

A team of engineers is conducting a land survey for a plot of land. The investigation aim at establishing numerical errors between a receivers indicated position and its true position using GPS (Global Positioning Satellite) primarily caused by atmospheric/time delays (x), satellite geometry (y) and elevation angle (θ) between the local horizontal plane(the ground) and the line-of-sight to the satellite. the relative position (P) in the GPS is modelled by the expression $P = xysin\theta$. The engineers will be certain of the position, if the relative error in P is given by $|\ \Delta\theta\ |\ cot\theta + \left|\frac{\Delta y}{y}\right| + \left|\frac{\Delta x}{x}\right|$ where $\Delta\theta, \Delta y\ and\ \Delta x$ are the errors in θ, y and x respectively. (Correct assumptions apply)

Satellite Geometry $y = tan\theta$ with $0 \leq \theta \leq 4\pi$. For correct location and sight of the plot, the satellite Geometry should have a percentage error in its approximation less than 1% using 6 ordinates

The time delay of signals from the satellite to the receiver results in a system of non-linear equation $2x^2 - 6x + 3 = 0$ that must be solved rapidly using appropriate iterations to pinpoint plot's coordinate, correct to 3 decimal places for $3 \leq x \leq 4$.

Task

(a) Help the engineers to be certain of the relative position of the plot, by analyzing the relative error.
(b) How correct is the location and sight of the plot of land with the help of the satellite Geometry?
(c) Predict the Pinpoint of the plots coordinate by analyzing the time delay of signals.

(WENSSEC)

ITEM 14

An entrepreneur has a problem of location of a lighting system on his compound. He consults a land surveyor who has a Global Positioning System (GPS) and finds out that the area is bounded by the line ($y = 10 - x$) and the curve ($y = e^x$) along the distances from $x = 2\ km$ to $x = 3km$. The land surveyor advised the entrepreneur that a graphical approach is the most suitable approximate method for location of a lighting system at the point of intersection between the line ($y = 10 - x$) and curve ($y = e^x$) along the distances from $x = 2\ to\ x = 3km$. the approximate point should be rounded off to 1 decimal place. But he would like to know the exact point using a suitable iterative formula to find the exact point of intersection. If the error between the two values is less than 0.05 then any of the two points can be used for installation of the lighting system.

The entrepreneur is also encountered with a problem of knowing the approximate area, he consults a land surveyor who finds out that the area is enclosed by the line ($y = 10 - x$), curve ($y = e^x$) and the distance from $2km\ to\ 3km$. One area was noted to be under the line ($y = 10 - x$) and another to be under the curve ($y = e^x$). He advised to use 0.05 sub-interval for the area under the curve $y = e^x$ and 0.1 subinterval of the area under the line $y = 10 - x$.There was a need to find the total approximate area using a suitable approximate method. There was also need to find the exact area using a suitable calculus method.

The land surveyor advised that if the approximate area and the exact area have an error less than 0.03, then any of the two areas can be used to estimate the area of the compound.

TASK

As a student of mathematics, help the architect to give a response to the businessman that has interpretation , methodology and judgment to the business

(SEC)

THEME 2: DATA ANALYSIS AND PROBABILITY

ITEM 1

A roadside vendor operating along the Kampala–Jinja Highway near Mabira Forest sells roasted chicken to travelers and commuters every day. To meet the daily demand, she stocks 100 pieces of chicken each morning. After keeping sales records for several weeks, she established the following probability distribution for the number of pieces sold each day:

Number of pieces of chicke sold	0	20	60	80	100
Probability odf selling	0.02	0.16	0.22	0.26	0.14

The vendor intends to use this information to estimate the expected number of pieces of chicken sold per day and the standard deviation of the daily sales. These statistics will help her plan future stock levels and minimize unnecessary losses due to unsold products.

Each piece of chicken is purchased from suppliers at UGX 3,000 and sold to customers at UGX 6,000. Since any unsold pieces are consumed by her family and therefore have no resale value, she wishes to determine her expected daily profit. If the expected daily profit is less than UGX 50,000, she intends to apply for a loan from a local money lender to increase her business capital and expand her daily stock.

To ensure that she sells high-quality products, the vendor purchases chicken from two wholesalers, A and B. The probability that she purchases her stock from wholesaler A is $\frac{1}{4}$, while the probability of obtaining good-quality chicken is $\frac{2}{3}$ when buying from wholesaler A and ⅓ when buying from wholesaler B. She wishes to determine the overall probability that a randomly selected piece of chicken from her stock is of good quality. If this probability is less than $\frac{2}{5}$, she plans to look for alternative suppliers with more reliable quality.

On one particular day, the vendor randomly selects five pieces of chicken from her stock for inspection. Assuming that each selected piece is equally likely to have been supplied by either wholesaler according to the stated probabilities, she wishes to determine the probability that exactly three of the five pieces were supplied by wholesaler A.

Task:

Help the vendor obtain the necessary information to guide informed decision making.

(UNNASE)

ITEM 2

A secondary school administration carried out an evaluation of students' performance in Economics to determine whether there had been any improvement in academic achievement over the course of the term. To do this, the administration compared the Beginning of Term (BOT) examination scores to students with their corresponding End of Term (EOT) examination scores. The results are shown in the table below:

BOT(X)	70	60	58	45	80	89	45	72	58	51
EOT(Y)	65	53	55	47	75	86	58	70	65	44

The administration wishes to illustrate the relationship between the BOT and EOT scores using an appropriate graph and determine whether a statistically significant relationship exists between the two sets of scores.

If the relationship is found to be significant at the 5% level of significance, the administration intends to organize an academic review meeting with the Economics teachers to discuss students' performance.

In addition, the Head of the Economics Department has been asked to prepare a statistical summary of the End of Term examination results for the entire class. The grouped frequency distribution of the EOT marks is presented below:

Marks	35– 45	−50	−< 60	−< 65	−< 75	−< 80	−< 95
Number of students	4	13	25	30	20	15	3

The administration would like to determine the modal mark for the End of Term examination. If the modal mark is less than 55, a meeting will be arranged with the students to discuss strategies for improving their academic performance.

Furthermore, the Head of the Economics Department has advised that the End of Term examination scores are approximately normally distributed. Based on this assumption, the administration wishes to identify the group of students who scored more than 85 marks, as these students will be appointed as leaders of the various Economics discussion groups.

Task:

Help the administration obtain the information needed to make informed decisions.

(UNNASE)

ITEM 3

A fitness coach at a fitness club conducted research to investigate the training outcomes of his clients. He found out that, on a given day, the average chance of a client using the chest press machine, the leg press machine, and the rope are 0.2, 0.3 and 0.6 respectively. The coach intends to recommend purchasing more equipment if the chance that a client uses at least two of the afore mentioned machines exceeds 65%. He then selected 12 of his clients to participate in an expansion study, on condition that the chance that at least 4 of these clients used the leg press machine on a given day exceeds 52%. Furthermore, he investigated the relationship between number of hours spent exercising and individual body weight. The data collected from 8 of his clients is as shown below;

Client	A	B	C	D	E	F	G	H
Hours spent exercising	1.0	1.2	1.5	1.8	1.9	2.0	2.2	2.4
Body weight (kg)	95	90	80	82	78	64	68	59

The club will increase the exercise hours for the clients if the research results show a correlation coefficient that is significant at the 1% level.

Task

(a) Make a recommendation regarding the purchase of more pieces of equipment for the fitness club.
(b) Decide whether there is a need to expand the training facility. (c) Make an informed decision regarding the hours of exercise.

(WAKISSHA)

ITEM 4

The management of an internet service company wants to improve the speed at which customers support calls are answered. The table below shows the number of customers who waited for their calls to be answered during a given day.

Waiting time (minutes)	Number of customers
0 - 4	8
5 - 9	14
10 - 14	22
15 - 19	26
20 - 24	18
25 - 29	10
30 - 34	7
35 - 45	5

The company wants to determine the most common waiting time. If this time is greater than 15 minutes, management will immediately recruit more customer service agents. Additionally, the data collected is considered realistic if the standard deviation is at most 11 minutes. The collected data shows that these waiting times follow a normal distribution. The board also wants to determine the 95% confidence interval for the mean waiting time. If the confidence interval lies entirely between 10 and 18 minutes, the customer service department will receive a pay rise.

Task

Based on appropriate mathematical calculations and justifications, advise the management on whether to:

(a) Recruit more customer service agents.

(b) Treat the data as realistic.

(c) Make a decision regarding the pay rise.

(WAKISSHA)

ITEM 5

A manufacturing company partnered with a nearby secondary school to improve efficiency in its community outreach programme. During a general cleaning exercise, the time taken by learners to complete an assignment task was recorded.

At the same time, the company monitored the quality of products manufactured by four machines (A, B, C and D). In addition, traffic flow on the road leading to the company was studied to help plan transportation for workers.

The following information was obtained:

The times taken by learners to complete the activity recorded as follows:

Time (seconds)	5-	10-	15-	20-	30-	50-100
Number of learners	35	42	58	14	3	1

In order to report effectively the company required the average time, the variance from the mean and the most frequent time by the use of an appropriate statistical diagram.

The manufacturing company uses four machines to produce its products. Machine A produces 30% of the daily output, machine B produces 30%, machine C produce 15% and machine D produce 25%.

The percentages of defective products are: Machine A 1%, Machine B 2%, Machine C 3% and Machine D 5%. In order to work on customer satisfaction, the company wanted the probability of product being defective and also that machine A would be used more to produce the products.

Between 8:00am and 9:00am, 40% of the vehicles travelling on the road move from west to east, while the remaining travel from east to west. A random sample of 250 vehicles is observed.

Task.

(a) Help the company to find the statistical measures of time for informed decision making.
(b) If a product selected at random is found to be defective, determine the probability that it was produced by machine A.

(JJEB)

ITEM 6

A meteorological agency is studying rainfall patterns and student performance.

In a simple model of the weather in July each day is classified as either fine or rainy. The probability that a fine day is followed by a fine day is 0.8. The probability that a rainy day is followed by a fine day is 0.4. The probability that 1 July is fine is 0.75.

The meteorologist wanted to discover whether 2 July would be fine and also 3 July could be fine.

At the same time, the amount of rainfall (in cm) on a randomly selected day is modeled by the continuous random variable X with probability density function

$$f(x) = \begin{cases} 3\alpha x & 1 \leq x \leq 3 \\ \beta(6-x) & 3 \leq x \leq 4 \\ 0, & otherwise \end{cases}$$

Where α and β are constants.

To assess academic performance, a random sample of 250 candidates is selected from the district. The sample has a mean examination score of 68.4 and a population variance of 182.25.

Task

(a) (i) Help the meteorologist investigate probability that 2 July will be a fine day then the probability that 3 July is also a fine day.
(ii) Investigate the probability that 3 July is rainy, given 1 July is fine.
(b) Help to construct a cumulative distribution function X, and hence determine the median for the population.
(c) Investigate the standard error and construct a 98.4% confidence interval for the population mean.

(JJEB)

ITEM 7

In the bid to improve the performance of mathematics, the head of curriculum in a certain school wants to analyze the performance of students in senior six. He wishes to know the mean and standard deviation. He also wishes to graphically show the distribution of the performance from which he will estimate the most likely mark of a particular test, students had performed as shown in the table below;

Marks	30–< 40	40–< 50	50–< 55	55–< 60	60–< 70	70–< 90	90–< 100
Number of students	2	15	19	22	18	17	3

He wishes that if the difference between the mean and the mode is less than the standard deviation, then a facilitation class will be arranged only for the challenged students.

He notices that the general performance of students in Mathematics is approximately normally distributed with mean 60.65 and standard deviation of 13.3. He wants to determine whether the percentage of students who scored above 75 is less the percentage of students who scored below 45 so that he can arrange for a retooling seminar for the teachers of the subject.

The probability of picking a football player from the mathematics class is 0.2. If a student is from the football player, the probability that they get D1 is 0.6. For students on the football team, the probability of getting D1 is 0.65. The head of the curriculum wants to find the probability that a math student is a football player given that they received a D1. If this probability is less than 0.5, extra Math lessons will be arranged for the football players.

Task:

Help the head of curriculum to prepare the information he needs for his analysis and make informed decisions.

(ACEITEKA)

ITEM 8

From the recently concluded mock examinations, 20 students of a certain school were not satisfied with their scores.

As a concern, the director examinations picked 8 of these students' scripts for a remark to determine the effectiveness of the mock examiners of this subject. The following table gives the scores for the 8 scripts:

Original mark	65	65	70	75	75	80	85	85
New mark	50	55	58	55	58	61	65	65

The director examination wants to verify if the two results correlate by using a suitable statistical diagram. The director examination wishes to write to the mock body to address the challenge in case the correlation in the two results is not significant at 1% level of significance.

The secretary to the director examinations recorded the scores of the twenty students using the original results but failed to clearly show the proportion of those who had scored 85 as follows;

Original mark	65	68	70	72	75	80	83	85
Proportion of students	0.05	0.15	0.1	0.22	0.15	0.05	0.1	a

The director examination wants to know the expectation and variance of these marks and also represent this data graphically for easy identification of the mode to be used in his write up to the mock body. He wishes that the difference in the expectation and the mode is less than the standard deviation to recommend the mock body examination for coming mock examinations.

The director examination also has established that the probability of one of these 20 students scoring a distinction in the final examinations is 0.65. He wants to know the percentage of exactly 15 of students scoring a distinction in their finals. If this percentage is less than 30%, then he will have to arrange for facilitation for the class.

Task:

Help the director of examinations obtain the information he needs to make informed decisions.

(ACEITEKA)

ITEM 9

During an inter-university sports tournament, a survey was conducted to gauge the viewing habits of students regarding the two most popular sports: Football and Cricket. The survey results showed the following probabilities:

- 35% watched Football but not Cricket

- 10% watched Cricket but not Football
- 40% did not watch either of the sport

With the next inter-university tournament expected to attract 500 students, and assuming viewing habits remain unchanged, the organizing committee is currently facing a challenge. They need to estimate the number of students expected to watch Cricket, given that they do not watch Football such that if the number exceeds 50, they would prepare a separate court outside the university.

However, a sample of 138 supporters was selected from the previous tournament who watched football. The table below shows the number of supporters and the time taken (in minutes) to enter the stadium, measured from the start of the game until halftime.

$Time\ (minutes)$	$Number\ of\ supporters$
$0 < x < 10$	10
$10 < x < 15$	35
$15 < x < 20$	65
$20 < x < 30$	107
$30 < x < 35$	123
$35 < x < 45$	138

The committee wishes to know the number of supporters who will arrive before the most likely time. If this number exceeds 50, they intend to offer a free drink to promote the game.

After the tournament, a survey was conducted among the spectators. A sample of 20 spectators was taken. If the probability of finding at most 12 spectators watching football is less than 0.4, the university plans to invest in activities other than football.

Task:

(a) Evaluate whether the university will prepare a separate court outside and ascertain whether the committee will offer a free drink to promote football on.
(b) Justify whether the university is planning to invest in other activities other than football.

(UMTA)

ITEM 10

A school board is reviewing the effectiveness of their Mock Examinations as a predictor of final A-Level success. The Director of Studies (DOS) has selected a sample data for **11 students** from last year's performance in Mathematics to check on the relationship between Mock exams and final A-level grades:

Marks in mocks	76	41	78	59	14	29	61	86	32	64	51
Grades in A level	A	B	B	C	D	E	B	A	D	C	E

The DOS need to assess whether mock examination marks provide a true and reliable estimate of students' expected performance in the final A-Level exams.

However, a sample of 81 students is taken, the chance of getting student scoring above the mean mark is 0.4 but board defines good performance if the chance of getting at least half the number of students scoring above the mean mark is greater than 0.5.

Basing on the district mock results, the marks of the whole district were **normally distributed** with a mean of 45 and standard deviation 5. The board wants to know the average mark for a sample of 100 students who sat for same mock examination in their school since the chance of getting a student above the average mark is 0.46. If the average exceeds 48 the budget of academics will be increased in order to improve on the performance in school.

Task:

(a) Justify whether mock examination marks provide a true and reliable estimate of students' expected performance in the final A-Level exams.
(b) Ascertain whether performance was good basing on the board's definition.
(c) Make a recommendation on whether the Academics budget will be increased.

(UMTA)

ITEM 11

The manager of a food processing plant is conducting a dual audit on quality control and employee demographics due to a recent health advisory.

In the packaging department, three workers Aileen, Barbara, and Cathy handle the daily batch. Aileen packs 55% of the total volume, Barbara packs 30%, and Cathy packs the remaining 15%.

Quality control records show that the probability of a packet containing broken biscuits is 0.7 if packed by Aileen, 0.2 if packed by Barbara, and 0.1 if packed by Cathy. During a routine inspection, the manager randomly selects a packet from the finished batch and finds it contains broken biscuits. He suspects Aileen is responsible.

On the same day, as part of a health-monitoring directive to track demographic vulnerability to an ongoing disease outbreak, the manager records the exact ages of all 40 workers on the factory floor:

19.2, 21.0, 22.5, 24.8, 25.0, 26.3, 27.1, 28.5,

29.0, 30.2, 31.5, 32.8, 33.0, 34.2, 35.5, 37.0,

38.4, 39.0, 40.2, 41.6, 42.1, 43.5, 44.0, 45.8,

46.2, 47.9, 48.0, 49.5, 50.8, 52.0, 53.4, 55.0,

56.7, 58.0, 59.2, 60.5, 62.1, 63.8, 64.0, 65.0

Task

(a) Evaluate the manager's suspicion. Based on the evidence of the broken biscuits, how likely is it that Aileen is responsible?

(b) Provide the manager with the average age of his workforce and by graphical analysis estimate the age threshold below which the youngest 75% of the workforce falls.

(KASSEC)

ITEM 12

UMEME, a major electricity distributor, is evaluating the performance of their light bulbs and employee productivity to improve their operations.

They have two types of bulbs:

Type A: Expected to last around 1150 hours, with some variation.

Type B: Expected to last around 1900 hours, with some variation.

The company wants to determine:

For Type A bulbs, what percentage are expected to last more than 1200 hours? If this percentage is less than 50%, they plan to replace all Type A bulbs.

For Type B bulbs, out of 1000 units, how many are expected to last less than 1800 hours? If fewer than 500 are expected to fail early, they may decide to stop using Type B bulbs altogether.

Moreover, the company is interested in understanding whether drinking coffee influences the productivity of their engineers. They recorded the daily cups of coffee consumed and the number of lines of code written by two engineers over a month:

X	10	15	20	25	30	35
Y	68	60	55	48	38	32

Umeme will base on a calculation to know if there is a relationship between X and Y and if found its exceeding 0.75 at 5% level of significance then coffee will be served daily.

Task:

(a) Decide whether the company should replace Type A bulbs accordingly and calculate how many of the 1000 Type B bulbs will likely fail before 1800 hours. Also, advise if they should continue using this type.
(b) Recommend whether the company should serve coffee daily to improve employee output.

(KASSEC)

ITEM 13

A District Health Office is evaluating the performance of a new mosquito-net distribution programme to determine whether it should be extended to all government health centres. To support this decision, the District Health Office must evaluate the efficiency of the

distribution process, the expected level of household participation, and whether the observed waiting times are representative of the district.

During a pilot distribution exercise, the waiting times of 90 households were recorded as shown below:

Waiting Time (minutes)	Number of households
0 – 2	6
3 – 7	14
8 – 14	35
15 – 24	20
25 – 39	15

The district has set a service standard that the most common waiting time should not exceed 15 minutes. The waiting-time distribution will therefore be analysed using a histogram to estimate the most common waiting time. If the estimated most common waiting time does not exceed the service standard, the current distribution procedure will be maintained; otherwise, the procedure will be reviewed before the programme is extended.

Previous records show that each household independently collects its allocated mosquito net with probability 0.82.

One government health centre serves 500 households. The district will only extend the programme if there is more than a 90% probability that at least 400 households will collect their mosquito nets. Otherwise, additional community mobilisation and sensitisation will be carried out before the next distribution exercise.

To determine whether the waiting time observed during the pilot exercise is representative of the whole district, a random sample of 64 households is selected from different health centres. The sample gives a mean waiting time of 16.8 minutes and a standard deviation of 6 minutes. District policy requires a 91.86% confidence interval to be constructed for the true mean waiting time before any decision is made. The programme will only be extended if the entire confidence interval lies below 18 minutes, providing sufficient evidence that the district service standard is likely to be achieved. Otherwise, further improvements will be made and another evaluation conducted before implementation.

Task:

Using appropriate mathematical and statistical methods, advise the District Health Office on the effectiveness of the mosquito-net distribution programme and its suitability for extension to all government health centres.

MATIGO

ITEM 14

The District Education Office is reviewing the effectiveness of its Mathematics Challenge Competition, which is used to identify learners for an advanced Mathematics enrichment programme. Eight representative learners from different secondary schools participated in

the competition and later sat the district end-of-term Mathematics examination. Their competition scores and examination scores are shown below.

Learner	A	B	C	D	E	F	G	H
Competition score	3	5	6	7	6	8	10	12
Examination score	50	54	65	62	70	67	73	78

The competition will continue to be used for selecting learners into the enrichment programme only if the association between the competition scores and the examination scores is significant at the 5% level based on Kendall's rank correlation coefficient, and the estimated examination score for a learner who obtains a competition score of 9 is at least 70 marks. Otherwise, the learner selection criteria will be reviewed before the next intake.

To encourage participation in future competitions, the organisers conduct a lucky draw at the closing ceremony. A new box containing 10 discs, of which 5 are red, 3 are black and 2 are green, is used. In this lucky draw, a disc is selected at random and its colour is noted. The disc is then returned to the box together with two additional discs of the same colour, and a second disc is selected. The organisers consider the lucky draw to be sufficiently fair if the probability that the first disc selected was red, given that the second disc selected is black, is at least 0.40.

The organisers urge the expected number of green discs selected in the two draws to estimate the number of green prizes required for future competitions. If the expected number is at least 0.40, the current allocation will be maintained. Otherwise, additional green prizes will be budgeted for future competitions.

Task:

Using appropriate mathematical and statistical methods, advise the District Education Office on the effectiveness of the Mathematics Challenge Competition and the suitability of the lucky-draw arrangements for future competitions.

MATIGO

ITEM 15

An entrepreneur has two companies producing two types of beverages Pepsi Cola and Mountain Dew. He consults his business analyst over his companies. The business analyst finds out that there is one company located in Apac (A) and another in Bukedia (B). The data analyst finds out that the likelihood of customers buying from district A and B are in the ratio 2:1 and that the ratio in thousands produced by company A are 6 Pepsi Cola and 8 Mountain Dew and that produced in company B are in the ratio of 9 Pepsi Cola and 7 Mountain Dew.

The business analyst informed the entrepreneur that there was need to make 2 pickings and find out the probability that the beverages are of the same type and if it is above 0.4 then the companies would increase the production of beverages of the same type. He further noticed that there was need to get the probability that they are from A given that they are of the same type.

The business analysis advised that there was need to develop a probability distribution of Mountain Dew type and find out the mean of the distribution of Mountain Dew and if it is above 1, then the Mountain Dew section would employ more people. He advised there was need also to find the standard deviation of the distribution.

The data analyst found out that there were other 30 similar companies countrywide and taking the probability of success is that of the Bukedia Company, he advised that there was need to get the probability of at least 9 successful companies and if it is above 0.2, then there would be a bank opened specifically for the Bukedia Company.

TASK:

As a Mathematics student, interpret, show the methodology and make a valid justification.

ASSHU- ANKOLE

ITEM 16

The board of governors (B.O.G) of a school is encountered with a problem of analysis of scores of their students during the end of term one that were compiled by the Director of Studies (D.O.S) of their school that was as indicated below.

Scores	$5-<30$	$30-<40$	$40-<50$	$50-<60$	$60-<80$
Number of students	17	21	30	14	3

The board of governors contacted a statistician who advised them that there was need of a graphical representation to help in knowing the mode, he also advised that a calculation for average score and standard deviation was necessary. He advised the board of governors that if students on average scored an average score less than 45 then all parents would be called for an academic assembly.

When the data was analyzed by the statistician, it was found out that it was approximately normally distributed. He advised the board of governors that there was need to know the 95% confidence interval around the average score. And if it lies between 35 and 45, there would be no need to administer remedial tests.

The statistician selected a sample of 10 students, where the probability between 35 and 45 taken as a normal distribution would be considered as a probability of success and therefore there was need to find the probability that at most 2 students are successful. And if the probability was found to be less than 0.2 then the director of studies would administer a suitable punishment.

Task

As a mathematics student, interpret, show the methodology and give justification to the board of governors.

ASSHU ANKOLE

ITEM 17

In a secondary school, the issuance of school uniforms for senior ones depends on their heights. As they proceed in the successive classes their uniform sizes keep changing and new uniforms need to be issued again. It is the school policy that only when the initial mean height is at least **125 cm** and the inter-quartile range of the heights not exceeding **16 cm** is when new uniforms will be issued. The heights are as follows;

Height (cm)	< 100	< 110	< 120	< 130	< 140	< 150	< 160	< 170
Cumulative frequency	0	27	85	215	320	370	395	400

Sometimes the head teacher prefers storing the heights of new senior one students in a graphical format to easily establish the most common height initially on admission to guide on the amount of material to purchase as students progress in the next classes. Unfortunately he is not a mathematician.

On the other hand, when a student completes **S.4** and joins **S.5** at the same school, free school uniform sets will be given as a reward under a condition that the relationship between the first 7 best performed subjects in **MOCK** and **UCE** is at least high. A student obtained the following grades:

MOCK	A	B	A	D	E	D	B	E
UCE	C	B	C	E	E	C	A	D

Tasks:

As a mathematics student, if consulted, advise the school to establish:

a. Whether new uniforms will or will not be issued.
b. How the head teacher will store the data and also highlight for him the most likely height to enable the change uniform.
c. In your own view, do you think a set of new uniforms will be given to that student joining S.5 with the scores provided? *(Justify your responses based by calculations.)*

COUHEIA

ITEM 18

In Uganda, the cases of road accidents are increasing because of reckless crossing and over speeding of pedestrians and drivers respectively. Because of this, traffic lights have been mounted on most busy roads. A case in point is the Kampala northern bypass whereby at a given point the speeds of the cars are normally distributed. Police found out that 95% cars travelled at a speed less than 85 kmh^{-1} and 10% travelled at a speed less than 55 kmh^{-1}.

On a certain day, an operation by police was carried out on 5 cars to find out how often these accidents occur. They came to realize that the chance of an accident to occur at this point was twice that of an accident not to occur. The O.C traffic instructed that if the chance that at least 3 accidents occurring in a week exceed 0.49, new road humps will be constructed unless otherwise.

In the process, many cars were impounded and each car had its issues depending on the mechanical conditions found on it. These conditions followed a continuous random variable that was uniformly distributed in the interval a *to* b. The mechanical conditions for car A had a lower quartile of 5 and an upper quartile of 9. It is a policy by the police to imprison a driver if the mechanical issues of the car lie in the interval between 2 and 12.

Tasks

As a senior six student if consulted, come up with a clear report on the following backed by calculations:

(a) The average speed and proportion of cars that exceeded 50 kmh^{-1}.
(b) Whether new road humps will be constructed or not.
(c) Whether it was possible for the driver of car A to be imprisoned or not.

COUHEIA

ITEM 19

One of the prominent Schools in Uganda has organized a visitation day, but for the previous years, parents always complain of traffic jam along the way to the school, since many parents will be moving on that day. The school is planning to meet the parents of candidates but it requires the modal, average and median time such that majority of the parents can attend the meeting.

The table below shows the time taken by parents to travel to school during visitation day:

Time (minutes)	<5	<10	<15	<20	<25	<30	<35	<40	<45
Number of parents	28	45	78	143	280	340	375	395	400

The meeting will be delayed if the interquartile range is less than 10 minutes.

The school is willing to give T-shirts to parents who will arrive in less than 18 minutes, on condition that majority arrive in less than 23 minutes.

On this very day, some parents used personal car (C) while others used buses (B) and taxis (T), with the proportions 40%, 50% and 10% respectively. For those who used personal car, 9% arrived late, and those who used buses, 6% came late and 20% of those who boarded taxi arrived at school late.

Task

As a student of mathematics,

(a) Help the school in planning such that they can have a wonderful visitation day.
(b) Determine the proportion that the vehicle used is a bus given that it arrives late.

UTES

ITEM 20

The director of studies (DOS) in a certain school is complaining why students are performing well in chemistry than in mathematics. He was trying to compare marks of students in both subjects and the table below shows the marks obtained by 10 students.

Mathematics (x)	10	15	20	15	30	35	40	45	50	60
Chemistry (y)	15	20	35	40	35	50	55	40	55	60

More teachers will be recruited if there is negative correlation between the performance in the above subjects. And extra lessons shall be created if the correlation is insignificant at 5%.

The DOS is wondering what could be the marks obtained in chemistry if a student scores 25 in mathematics.

During printing of report cards, the secretary occasionally makes mistakes. In a batch of 20 cards, 5 are found to have mistake. A sample of 3 batches were at random and examined, the DOS wishes to know the expected number of mistakes and standard deviation made by the secretary.

Task

As a student of mathematics; help the Director of Studies clarify on the disparities of marks in the two subjects and the mistakes in order to make an informed decision.

UTES

ITEM 21

At a certain school, the games master Mr. Kasadha is preparing students for a district athletics and games competition. He measured the weights in kilograms of 40 students that are set to participate in the competitions and their weights were recorded as follows:

Weight (kg)	$30-<35$	$35-<40$	$40-<55$	$55-<60$	$60-<65$
Frequency	8	5	12	9	6

At the same time, Mr. Kasadha records a random sample of 100 race times x in seconds from students who were training for a given race. From the sample results, it was observed that $\Sigma x = 380 \; and \; \Sigma x^2 = 1840$ where x represents the race times recorded. According to the training standard requirements, all participants whose weight is heavier than the average weight need special fitness attention and the acceptable range of weights for the participants is the middle 70% of the weight distribution of the participants. Mr. Kasadha does not know the specific facts about the participants in line with the training standard requirements.

Mr. Kasadha has to assess whether the sample results recorded on the race times of the students training are precise enough before planning for further training sessions and to judge how precise the recorded results are, the width of the 90.8% confidence interval for the mean race times of all students who were training for the race must be at most 0.5. He wants to plan for extra training sessions to prepare for the district athletics and game competition.

From the past training records, the number of extra training sessions needed by a given student is represented by the random variable Y where Y $= 1, 2, 3,$ or 4 and the probability that a student needs y extra training sessions is given by:

$$P(X = x) = \begin{cases} \frac{1}{60}(ay + b), y = 1, 2, 3\ 4 \\ 0, \qquad elsewhere \end{cases}$$ Where a and b are constants.

It is also known that $F(3) = \frac{13}{20}$. Mr. Kasadha wishes to know the expected number of extra training sessions needed to prepare the students and the number of students who need more than one extra training session.

Task

Help Mr. Kasadha find the specific facts about the participants in line with the training standard requirements, assess the preciseness of the sample results, determine the expected number of extra training sessions and the number of students who need more than one extra training session.

MEC

ITEM 22

The school board of governors organized a meeting to assess and analyse feedback collected from parents, teachers and learners about academic performance, students' discipline and participation in co-curricular activities.

The school administrators reported that out of all feedback forms received 50% were about academic performance, 30% were about students' discipline and 20% were about co-curricular participation. Records further showed that 2% of the academic feedback forms raised serious concerns, 3% of the discipline feedback forms raised serious concerns and 4% of the co-curricular feedback forms raised serious concerns.

Given that an academic feedback form was picked at random during the meeting, the board wanted to know the chances that the form raised a serious concern.

As part of improving the school academic welfare, the board of governors interviewed seven applicants for the post of Deputy Headteacher.

The seven applicants labelled A, B, C, D, E, F and G who sat for the two separate sets of interviews, the technical interview and the leadership interview.

The applicants obtained the following positions according to performance in the two sets of interviews:

Position	1st	2nd	3rd	4th	5th	6th	7th
Technical interview	G	F	A	D	B	C	E
Leadership interview	D	F	E	B	G	C	A

The board wanted to know whether the two sets of interviews ranked the applicants significantly before deciding which applicants would proceed to the strategic presentation interview before selecting the successful applicant.

In addition, the headteacher reported that the scores in general paper obtained by the 280 candidates in S.6 in the recent set of exams were approximately normally distributed with 37.5% of the candidates scoring less than 40 and 12.5% scoring more than 60.

The board is concerned about the performance because it could affect their scores in the final exams and wish to know the percentage of candidates who were awarded a point for a score of 45 and above and the minimum score obtained by the top 10% of the candidates.

Task

As a mathematician, help the board of governors determine the chances that the form picked during the meeting raised a serious concern, whether the two sets of interviews ranked the applicants significantly and find the number of candidates who were awarded a point and the 80th percentile of the scores obtained by the candidates.

MEC

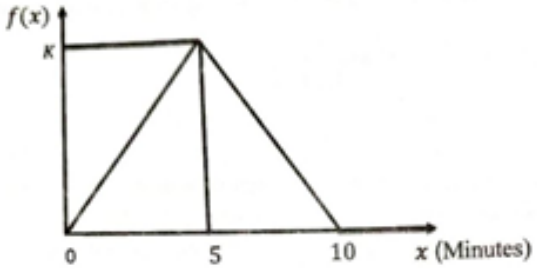
ITEM 23

A certain school is planning on strategies to improve students' performance so as to improve the school enrollment. The S.6 class teacher subjected a group of students to a number of tests and the average marks for the students were as shown in the table below:

Marks	Number of students
Below 10	6
10 *and under* 25	11
25 *and under* 35	18
35 *and under* 45	25
45 *and under* 50	15
50 *and under* 65	13
65 *and under* 80	8

The school administration decided that if the modal mark is less than 50, they should bring in external facilitators to boost the performance. If the middle 60% range of marks is less than 38, another stream should be created in the class.

The school has also organized a seminar for S.6 students scheduled to begin at exactly 8:00 am. In order to improve on punctuality and planning for future events, the teacher in the department collected data on the arrival times of participants. Arrival time was measured in minutes after 8:00 am and was represented by a continuous random variable X. The probability density function f(x) is as shown in the sketch graph below:



The teachers in the department need to know the constant value k and then express the equation of the probability density function f(x) in its complete algebraic form.

The following rules were set:

If the probability that a participant arrives within 4.3 minutes after 08:00 am is greater than 0.60, the current reporting system will be maintained.

If the 90th percentile arrival time is more than 7 minutes, measures will be taken to improve punctuality.

Task

Help the teachers to make an informed decision

TAEB

ITEM 24

The management committee of a secondary school became concerned about learners' academic performance, staff punctuality, and the safety of the school transport system after several complaints from parents and teachers. During a planning meeting, the committee resolved that future decisions should be guided by proper mathematical analysis rather than assumptions. They therefore wished to study examination performance in order to improve academic standards, evaluate the reliability of staff punctuality arrangements, and assess whether the road commonly used by the school van was safe enough for regular transport activities.

The marks obtained by learners in a mock examination were grouped as shown below:

Marks	Number of students
$0 < x \leq 15$	6
$15 < x \leq 30$	8
$30 < x \leq 35$	15
$35 < x \leq 40$	20
$40 < x \leq 55$	10
$55 < x \leq 65$	7
$65 < x \leq 75$	4

The Committee wished the Director of Studies to estimate the median mark using a suitable statistical graph and also use a computational method to determine the mean mark and

standard deviation. The committee will maintain the current teaching plan if the mean mark is at least 50%; otherwise, weekly tests will be introduced.

The committee also studied the punctuality of a staff member who travels by either murram road or tarmac road. The chance that he uses murram road is 0.25. If murram road is used, the chance that he arrives late is 0.6, while for tarmac road it is 0.3. The committee will keep the current staff travel arrangement if the chance that the staff member used the tarmac road, given that he arrived late, is less than 0.7; otherwise, a stricter punctuality plan will be introduced.

In addition, accident records on a road used by the school van showed that the monthly number of accidents is uniformly distributed from 5 to 12. The committee wishes to determine the expected number of accidents in a month and the chance of having more than 7 accidents. The school will continue using the route for regular transport if this chance is less than 0.4; otherwise, an alternative route will be chosen.

Task

As a student of Mathematics, help the school management committee interpret the information, choose suitable methods, and make justified decisions about academic support, staff punctuality, and school transport.

TAEB

ITEM 25

The ministry of health is analyzing the efficiency, timely delivery of essential drugs by National medical stores (NMS) and cases of stock outs and expiry of drugs to different hospitals in the country.

The table below shows the delivery times (in hours) to different hospitals:

Delivery times (in hours)	$5-<10$	$10-<20$	$20-<30$	$30-<40$	$40-<60$
Number of hospitals	10	15	25	40	25

For timely delivery of essential drugs, NMS must have coefficient of variation less than 50% (indicating a lower degree of variation or more consistent delivery of drugs to hospitals).

Two drivers of NMS, John and William used the same route to deliver medicines to Arua Hospital. John is thrice likely to be used to deliver medicine as William. The chance for the drivers to be timely is 0.65 for John and 0.3 for William. The efficiency of the drivers would not cause stock out of medicines and cases of expiry in Arua Hospital if the chance for John to be used given that the drivers are timely should be greater than 0.7.

The Ministry's bio statistician is expected to analyze the efficient delivery schedule for 30 different Hospitals given that; Average delivery time is 6.4 days with consistency of 2.84 days. Basing on 90% assurance,

Task

(a) Is NMS timely in delivery of essential drugs to hospitals? Justify.
(b) Analyze the efficiency of drivers to ascertain whether there are likely cases of expiry and stock out of medicines.
(c) Provide the most efficient delivery schedule by analyzing the Bio statistician's data

WENSSEC

ITEM 26

Arua city authorities are investigating the cases of crime committed by students during ballgames. The table below shows the record of crimes at the central police station by students of different age groups in 2026.

Age of students (years)	10–14	15–18	19–21	22–24	25–29	30–34
Number of crime cases recorded	9	28	26	34	6	2

Previous year (2025) record indicates that the cumulative trends in crime cases that included the spread of middle 50% range of the crime cases were 18 years old students. The RCC (Resident City Commissioner) will suspend the ball games if the record of 2026 shows an increase in the age of learners who are involved in crime cases, indicating involvement of mature students and/or old students.

Three Umpires; 1 (Chief), 2 and 3 officiated Netball 2025 Ballgames. The table below shows the points awarded to ten different schools by the three Umpires.

School	A	B	C	D	E	F	G	H	I	J
Umpire 1	72	50	50	55	35	38	82	72	60	45
Umpire 2	60	55	56	50	50	50	73	70	58	51
Umpire 3	50	40	67	70	67	48	67	48	40	62

The organizing committee wants to appoint one Umpire to officiate 2026 ballgames together with the Chief Umpire since few schools turned up for the ballgames in fear of suspension of games by the City Authorities.

An Umpire will be appointed if the performance of the Umpire in 2025 with the Chief Umpire is significant at 1% level.

During the Finals of 2025 Football, Team G qualified to play team H. They drew and five penalties each were awarded to the teams. The players of team G and H had 0.45 and 0.5 chances of scoring a penalty respectively. A Team will be declared a winner if the chance of them scoring at least 3 of the 5 penalties is greater than 0.5.

Tasks

(a) Analyze the crime cases to ascertain whether the ballgames were suspended or not.
(b) Recommend the two Umpires that needed to be appointed to officiate ballgames in 2026. Justify.
(c) Which team won the Football of 2025? Justify your response.

WENSSEC

ITEM 27

A Board of Governors (BOG) of a school is planning to analyse the mock results of Principal Mathematics (P425/2). The Director of Studies (DOS) is requested to provide the scores and their respective number of students that were as indicated in the data below.

Scores	No. of Students
$0-<5$	44
$5-<15$	81
$15-<30$	105
$30-<50$	146
$50-<70$	98
$70-<90$	46

The Board of Governors contacted its statistician who advised that there was need to find the average score of their students. If it was found to be less than 45, they would call an academic assembly. He also advised that there was need of a graphical approach from which to state the modal age group.

The statistician found that the data provided was normally distributed. He advised that there was need to find out the unbiased estimate of variance and therefore the standard deviation that would be used in estimation. A sample of 40 students was selected among the given ones.

The probability of success is that of scoring at least 50. The statistician advised that there was need to get the probability of most 7 successful students. If it is found to be less than 0.4, then the DOS will administer a punishment.

The statistician advised the GOG that there is need to find the confidence interval around the average score. If the interval from 35.45 lies within the interval got, there would be no need to administer remedial tests

Task

As a Mathematics student, interpret, use a suitable methodology and give a justification to the board of governors over the scores and numbers of students provided by the DOS.

SEC

ITEM 28

The academic committee of Parents Teachers Association (PTA) is concerned about the performance of their Biology and Economics class in a post Mock Examination. They contacted the Director of Studies (DOS) who provided the scores as below.

Candidate	Biology	Economics
A	45	90
B	65	76
C	61	70

D	83	75
E	15	50
F	77	85
G	70	33
H	62	62

The PTA presented the data to a data analyst. The data analyst advised that there was need to represent the data graphically to find out the relationship between Biology and Economics.

The data analyst also calculated the correlation coefficient for the two subjects as one sitting. Students offering the two subjects would have need to have correlation coefficient to be greater than 0.75.

The data analyst selected three students A, B and C and found out that the probability of passing Biology was $\frac{3}{5}, \frac{1}{5}\ and\ \frac{2}{5}$ respectively.

He requested that data was needed to formulate a probability distribution of passing Biology by 3 students. Go ahead and find out the average number of passing biology. If it was found to be greater than one, then there would be need to admit more students of Biology, the standard deviation is also important in the analysis.

The data analysis found out that in a sample of ten students above, the probability of passing Biology was found to be $\frac{2}{3}$. He advised that if it was a need to find the probability that at most 2 students passed Biology; if it was found to be greater than 0.3, there would be need to reward students for their performance.

TASK

As a mathematics student, analyse, interpret and justify to the PTA over the performance of their students in the two subjects.

SEC

THEME 3: MECHANICS

ITEM 1

A football club is conducting a pre-season fitness assessment to prepare its players for an upcoming tournament. As part of the strength-training programme, one of the players, who has a mass of 80 kg, performs a suspension exercise by hanging motionless from two training ropes attached to an overhead support. The ropes are inclined at 20° and 40° to the vertical.

According to the manufacturer's safety specifications, each rope can safely withstand a maximum tension of 300 N. Before continuing with the exercise, the coaching staff wishes to determine the tension in each rope and decide whether the ropes are safe for use or should be replaced. During the same training session, the player practices shooting accuracy by kicking a football from level ground with an initial speed of 30 m s^{-1}. He aims to land the ball at a teammate positioned 45 m away on the same horizontal pitch. To improve his shooting technique, the coach asks him to determine the possible angles of projection that will enable the ball to reach the target.

To conclude the training session, the player completes an endurance drill by running two consecutive distances of 500 m. He covers the first 500 m in 1 minute and the second 500 m in 3 minutes. Assuming that his speed decreases with uniform retardation throughout the drill, the coach wishes to estimate the additional distance the player would run before eventually coming to rest.

Task

Help the coaching staff analyse the player's training performance and make informed decisions.

UNNASE

ITEM 2

The Police Traffic Control Unit is monitoring vehicle movement along the busy Kampala–Jinja Highway due to a recent increase in road accidents. To enhance safety, a real-time vector-based tracking system has been installed to monitor positions and velocities of moving vehicles in three-dimensional space.

On a particular day:

A trailer departs from Kampala at $11{:}00\ a.m$, position $(2i - 3j + 4k)\ km$ with constant velocity $(2i + 2j + 3k)\ km/h$

A bus departs from Jinja at $12{:}00\ noon$ from position $(7i + 5j - 2k)\ km$ with constant velocity $(-3i - 8j + 18k)\ km/h$

Due to the high traffic density, the control center must determine whether the two vehicles are on a collision course and, if so, identify the exact point of collision in order to dispatch emergency services.

According to new traffic regulations, all accident vehicles must be towed separately to a nearby police station for forensic investigation.

In one such incident, a trailer of mass 5 tonnes is towed using a recovery vehicle of mass 1.5 tonnes with a driving force of 60 kN up a rough road inclined at 3 in 5. The coefficient of friction between the vehicles and the road is 0.25.

Tasks

(a) Use the information above to determine whether a collision occurs and find the time and position of the collision.
(b) Calculate the tension in the tow bar between the trailer and the recovery vehicle.

UNNASE

ITEM 3

An officer working with Uganda National Roads Authority (UNRA) is responsible for ensuring safety on Uganda's major highways. One morning, the traffic control centre received information that two fuel tankers are approaching the same section of the Kampala–Jinja highway, a busy route used to transport petroleum products from Jinja to Kampala.

At 9: 00 $a.m$, truck A is at position (0,0) km and is travelling east at a constant speed of 60 $km\ h^{-1}$. At the same time, truck B is at position (15,8) km relative to the fixed point and is travelling south at a constant speed of 48 $km\ h^{-1}$.

The road safety department requires an emergency warning to be issued by the engineer whenever two vehicles are expected to come within 2 km of each other.

After arriving safely at a fuel depot, truck B must unload 4 tonnes of packaged lubricants packed in 200 kg batches. One batch at a time is pushed up an 8 m smooth ramp inclined at 25° to the horizontal, using a constant force of 1000 N parallel to the incline. The depot manager has allowed 2 minutes to complete the unloading before the next delivery truck arrives.

Task

(a) Establish whether the emergency warning should be issued by the engineer.
(b) Confirm whether the unloading process of truck B will interrupt the next delivery.

WAKISSHA

ITEM 4

A telecommunications company is installing a new signal mast in a rural area of Uganda. The mast is held in equilibrium by five supporting steel cables attached at different points, pulling in different directions simultaneously.

The forces in the cables are summarized below:

Cable	1	2	3	4	5

Force (kN)	14	9	18	11	8
Direction	$Due\ East$	$N40°E$	$Due\ West$	$Due\ South$	$S30°E$

Safety regulators state that if the resultant force on the mast exceeds 12 kN, the structure is considered unstable and must be reinforced immediately.

After installation, a fuel truck must ascend a steep access road leading to the site. The steepest section of the road is inclined at 12° to the horizontal. To ensure continuous motion uphill, the tanker's engine provides a constant driving force of $P = 42\ kN$. The coefficient of friction between the tyres and the road surface is $\mu = 0.25$. The steep section of the road is 250 m long, and the tanker enters it with an initial speed of 6 m s^{-1}. The tanker and its load have a total mass of 1800 kg. The vehicle's transmission system is designed to safely operate up to a maximum speed of 100 m s^{-1} under a heavy load. If this limit is exceeded, there is a risk of mechanical failure, which could cause the tanker to lose control on the steep section.

At the same time, a worker operating a crane accidentally dislodges a metal tool from a platform at the top of a cliff, which is 50 m above the road. The tool leaves the platform with an initial speed of 20 $m\ s^{-1}$ at an angle of 35° above the horizontal, directed towards the road where the tanker is moving. The horizontal distance from the base of the cliff to the tanker's entry point on the road is 45 m.

Task

As a site engineer:

(a) Establish whether there is a need for structural reinforcement.
(b) Determine whether the tank will ascend without a mechanical risk.
(c) Analyse whether the falling tool can cause danger to the tanker.

WAKISSHA

ITEM 5

The Uganda Civil Aviation Authority (UCAA) is conducting a search and rescue training exercise on a lake. The exercise involves a rescue helicopter, two emergency response vehicles, and two rescue motor boats.

During the exercise, the helicopter launches a rescue package from a cliff with an initial speed of 30 m s^{-1} towards a target located 60 m horizontally from the point of projection and at the same level. For safety reasons, the launch angle must be between 20° and 67°. At the same time, two emergency response vehicles leave from opposite ends, A and B, of a 20 m straight road. At the same instant, one vehicle starts from rest at A and travels towards B with a constant acceleration of 2 m s^{-2}, and the other vehicle starts from rest at B and travels towards A with a constant acceleration of 5 m s^{-2}. The intersection occurs where the vehicles meet.

Meanwhile, two rescue motor boats P and Q begin moving on the lake. Boat P travels at a constant speed of 20 km h^{-1} on a bearing of 060°, while boat Q travels at a constant speed of

10 km h^{-1}. At 2:00 pm, the operator on boat Q observes that boat P is 4 km due west of Q. The intention of the operator on Q is to pass as close as possible to P.

Task

(a) Investigate whether the rescue was successful.
(b) Investigate the point on the road where the particle would meet from A.
(c) (i) Investigate the course that must be set by boat Q to pass as close as possible to P and the shortest distance.
(ii) Help the operators to know the time when the rescue will take place.

JJEB

ITEM 6

An engineering team is transporting equipment to a construction site across a river. The equipment is first pulled up a rough inclined plane with an angle of inclination $\sin^{-1}\frac{3}{5}$. The coefficient of friction between P and the plane is $\frac{1}{8}$. The equipment is pulled by a light inextensible string passing over a smooth pulley. A 5 kg mass on the incline is connected to a hanging load Q of 4 kg. Q is initially 2 m above the ground. The system is released from rest with the string taut.

At the top of the incline, four guide cables exert forces of $4\ N, 5\ N, 9\ N, and\ 6\sqrt{2}N$ along AB, CB, AD, and AC respectively, where ABCD is a rectangle and $AB\ =\ 2\ AD$. The engineers wish to replace these four forces with a single equivalent force.

After unloading the equipment, the team crosses a river 50 m wide flowing at a constant speed of 3 m s^{-1}. Their boat has a speed of 2.5 m s^{-1} in still water. The boat leaves a point on one bank to reach a point 35 m downstream from its starting point. The crew had two possible courses in crossing the river, and the best would be where least time is taken.

Task

(a) Investigate the acceleration of the system and the time taken for the hanging mass Q to reach the ground and the distance mass P moves up the inclined plane.
(b) Investigate the single force that could replace the four forces.
(c) Advise the crew on the course to be taken.

JJEB

ITEM 7

A newly trained pilot flying from airfield A to airfield B faces three distinct challenges: managing stability of a cargo, navigating safely to avoid mid-air collision, and executing a precise projectile release.

He plans to fly an aircraft from airfield A to airfield B, carrying a 10 kg cargo attached by an inelastic string. During the flight, a constant horizontal force of 50 N acts on the cargo in the

opposite direction of motion, causing the string to displace from the vertical by a certain angle. To ensure a safe flight, the pilot needs to determine this angle of displacement and calculate the tension in the string, knowing that if the tension exceeds 120 N, he will be forced to double the strength of the string.

At 8:00 AM, the pilot sets off from airfield A at a velocity of 110 km h^{-1} in a direction of 320°. Airfield B is located 32 km away from airfield A in a direction of 250°. Simultaneously, a second aircraft departs from airfield B, flying at the same altitude with a velocity of 104 km h^{-1} in a direction of 030°. To avoid a collision, the pilot wants to determine the shortest distance between the two aircraft during their motion and the exact time it occurs. If this shortest distance is less than 10 km, he will have to adjust his flight altitude.

Once past the point of closest approach, the pilot intends to release the cargo at a point when flying at a height of 250 m above ground and when moving at a velocity of 110 km h^{-1}. He must release the cargo so that it undergoes projectile motion and lands precisely at a target point C. The ground distance from the point of release to point C must not exceed 220 m for the cargo to remain within a recoverable distance.

Task

Help the pilot get clear information on this flight and make informed decisions before he sets off.

ACEITEKA

ITEM 8

The traffic control office is monitoring two motorcycle ambulance riders who are responding to calls from different parts of a town in Eastern Uganda. The town centre roundabout is taken as the origin on a coordinate grid. Distances are measured in kilometres and speeds in kilometres per hour. At exactly 10:00 am, the control screen gives the following information:

For rider A: $R_A = \begin{pmatrix} 4 \\ -2 \end{pmatrix} km$, $V_A = \begin{pmatrix} -12 \\ 12 \end{pmatrix} km/h$

For rider B: $R_B = \begin{pmatrix} 2 \\ 2 \end{pmatrix} km$, $V_B = \begin{pmatrix} -6 \\ 0 \end{pmatrix} km/h$

The control officer must decide whether the two riders are on a collision course. The situation is dangerous only if the riders would reach the same position at the same time within 30 minutes if they maintain their directions or speed. If the situation is dangerous, the officer must warn the riders.

Later, a community tractor breaks down while carrying sacks of maize. A pickup truck is used to tow it along a rough road to a safe parking yard. Before towing begins, the local mechanic must check whether the operation is safe. The following data is provided.

Quantity	Value
Mass of broken tractor	1500 kg

Mass of pickup truck	1700 kg
Driving force of pickup	7200 N
Coefficient of friction for tractor	$\mu = 0.18$
Road resistance on pickup	950 N
Gravitational acceleration	$g = 9.8\, m\, s^{-2}$
Connection	Light inextensible chain

The towing is approved only if the tension in the chain does not exceed 4350 N.

The ambulance riders have completely run out of critical medical supplies and have to fetch them from a nearby trading centre. The trading centre is separated from the town by a fast-flowing river of width 0.8 km, moving at 5 km/h to the east. Rider A is selected to pilot a speedboat capable of travelling at a constant speed of 15 km/h to fetch supplies from the trading centre. Rider A departs from port P to reach a port Q, which is located 0.8 km downstream of point P on the opposite bank. After loading the supplies, Rider A must return to point P. However, the boat's propulsion battery is low, and the speedboat's digital navigation display runs entirely on a small battery. At the start of the journey, it indicates only 12 minutes of power remaining.

Tasks

(a) Help decide whether:
  i) The ambulance riders should be warned.
  ii) The towing operation should be approved.
(b) Determine whether Rider A will return to point P before the battery shuts down.

ACEITEKA

ITEM 9

A civil engineering firm is analysing the forces acting on a regular, two-dimensional steel truss component used in a pedestrian bridge design. The truss structure is shaped like a regular hexagon, ABCDEF, with each side measuring 2 m and has a mass of 500 kg.

The structure is subjected to eight internal forces due to its formation. These forces and their directions, indicated by the order of the letters of the truss, are as follows: $9\ N, 5\ N, 7\ N, 3\ N, 1\ N, 3\ N, 5\ N\ and\ 4\ N$ act in $AB, BC, CD, DE, EF, FA, AD\ and\ AC$, where AB and AE are in the x- and y-planes respectively.

The engineering team needs a complete vector analysis to ensure the truss can withstand the resultant load of 12 N.

However, in the process of lifting the truss to the bridge using a crane, the rope snapped and the truss fell from a height of 5 m vertically onto rough ground inclined at 25° to the horizontal, with a coefficient of friction of 0.3. It slides using an initial speed of 7 m s^{-1} for a distance of 8 m and hits a vertical wall. The team wishes to know the speed at which the truss hits the wall.

Engineers assert that for the truss to remain intact, it should not hit the ground with a velocity greater than 2.5 m s^{-1}, or else it will be deformed.

Task

(a) Determine whether the truss can withstand the resultant load and its direction of action.
(b) Justify whether the truss will remain in its form and the speed with which it hits the wall.

UMTA

ITEM 10

On a certain island around Lake Victoria, smugglers are loading one tonne of fuel containers onto a boat using a rough wooden ramp. The ramp is 10 m long and inclined at 25° to the horizontal, with a coefficient of friction of 0.1. Each container, with a mass of 100 kg, is pushed up towards the boat by a horizontal force of 700 N. The smugglers are uncertain about how much time it will take to load the containers, but they have a strict time window of only 2 minutes to complete their work.

Just after midnight, a marine patrol officer was patrolling the waters near Entebbe. At 1:00 AM, his spotter noticed a suspicious motorized boat believed to be smuggling fuel and untaxed goods that was 20 km due East of their position and heading due N15°E at 15 m/s. The smugglers were attempting to reach the safety of a protected island by 2:00 AM. The officer is not sure of the time it will take to intercept them, since the top speed of his boat is 20 m/s.

Task

(a) Justify whether the smugglers will be able to complete their work in time.
(b) Is the patrol officer able to intercept the suspicious motorboat in time?

UMTA

ITEM 11

A construction team at a logistics warehouse is setting up an automated roof system. A crate of machinery ($Body\ A$, $m_1 = 20\ kg$) sits on a rough inclined roof tilted at an angle of $\theta = 30°$ above the horizontal. This crate is connected by a light, inextensible cable passing over a frictionless pulley to a heavy counterweight ($Body\ B$, $m_2 = 15\ kg$) that hangs freely vertically.

The coefficient of kinetic friction between Body A and the roof is $\mu_k = 0.15$. When the system is released from rest, the hanging counterweight pulls Body A up the incline.

The resultant force acting on each of the two bodies must be at least 140 N, the tension in the cable more than 135 N, and the rate of change of velocity should not be less than 0.5 m s^{-2} if they are to continue with construction work.

A war missile ($mass = 5000\,kg$) is stuck in a harbor channel. Two towboats tie cables to the front of the missile to pull it into the docks.

Towboat A pulls with a force of $F_1 = 1500\,N$ directed at an angle of 30° North of East.

Towboat B pulls with a force of $F_2 = 1200\,N$ directed at an angle of 45° South of East.

Due to the narrow canal, there is a constant water drag (friction) resisting the forward motion with a force of $F_{drag} = 400\,N$ directly West.

Task

Help the construction team to:

(a) Determine whether the crate–plane goals above are achievable.
(b) Discover what single force is required and the direction it should take in order to take out the missile from water.

KASSEC

ITEM 12

The lake authorities have received reports of illegal fishing activities in a remote part of Lake Bunyonyi. To catch the offenders, a police patrol boat (Boat P) is dispatched from the main dock in position A, given by $(3i + 8j)$. The patrol boat travels at a steady speed of $20\ km\ h^{-1}$ along a straight route heading 45° north of east.

Meanwhile, an illegal fishing boat (Boat F) has been spotted at position B, given by $(8i + 6j)$. The fishermen are trying to reach a protected area whose position is C $(12i + 10j)$. They are moving at a steady speed of $15\ km\ h^{-1}$ directly towards point C.

The police are aiming to intercept the fishing boat before it reaches the protected area to prevent illegal fishing activities.

The authorities need to determine:

- When and where the police boat can intercept the fishing boat.
- Whether the police can catch the fishing boat before it reaches point C.

Task

Help the authorities determine the time at which the police boat is closest to the fishing boat during their journeys, whether the police can intercept the fishing boat before it reaches point C, and if possible, coordinates where the police boat should meet the fishing boat.

KASSEC

ITEM 13

The Ministry of Works and Transport is investigating the safety of a steep section of a certain road during dry and rainy seasons for heavy trucks. The section of road is inclined at an angle

of 12° to the horizontal, and the estimated road section is determined at an angle of 12° to the horizontal.

Coefficients of static friction between the tyres of heavy trucks and the road surface are as shown below:

Road condition	Coefficient of friction
Dry season	0.30
Rainy season	0.12

As part of the investigation, a fully loaded supply truck of mass 15 tonnes is positioned on the slope and is initially at rest with brakes engaged and engine switched off. The truck's parking brake is then released. If the truck remains stationary after the parking brake is released, the road will remain open to heavy vehicles during that season. Otherwise, a second safety test will be carried out to determine whether disabled heavy vehicles can be recovered safely.

For the second test, a breakdown truck of mass 20 tonnes, having a maximum engine driving force of 120 kN, is connected to the supply truck using a tow cable with a maximum safe working tension of 50 kN. If the breakdown truck is able to pull the supply truck up the slope without the tension in the tow cable exceeding its maximum safe working tension, the road will remain open to heavy vehicles during that season, provided that a fully equipped breakdown truck is kept on standby at the site. Otherwise, the road will be temporarily closed to heavy vehicles during that season until a higher-capacity cable or more effective recovery arrangements are put in place.

Hint:

Assume that:

- Both trucks are on the same 12° incline.
- The tow cable is light, inextensible, and remains parallel to the road surface.
- The coefficient of friction between the tyres and the road remains constant for each road condition throughout the investigation.
- Rolling resistance and air resistance are negligible.
- Acceleration due to gravity, $g = 9.8\ m\ s^{-2}$.

Task

Using appropriate principles of mechanics and mathematical justifications, advise the Ministry of Works and Transport on the appropriate measures for the safe operation and recovery of heavy vehicles along this section during the dry and rainy seasons.

MATIGO

ITEM14

An emergency response agency is coordinating an air-and-ground relief operation to deliver medical supplies to a flood-affected area.

Two aircraft fly at the same altitude. Aircraft A flies at a constant speed of 720 km h^{-1} on a bearing of 045°, whilst Aircraft B flies at 540 km h^{-1} on a bearing of 030°. Initially, Aircraft B is 100 km from Aircraft A on a bearing of 065°.

The aircraft will remain under airport radar surveillance for only the first 6 minutes after departure. Airport surveillance covers the first stage of the mission, after which the aircraft will proceed to continue flight into the zone where airport radar surveillance is no longer available. Otherwise, the airport radar will provide sufficient information.

In addition, safety regulations require the aircraft separation to be a minimum of 20 km. The minimum separation is less than 30 km, otherwise at least 10 minutes are required after flight before departure. Otherwise, the planned flight paths are approved.

After landing, a worker launches a medical package from a stationary platform 1.2 m above the ground at a speed of $15\sqrt{2}\ m\ s^{-1}$ at an angle of 45° to the horizontal. At the same instant, a supply truck with a constant speed of 10 m s^{-1} moves directly away from the platform in a horizontal direction. A worker aims the package so that it lands in the truck bed, which is 1.2 m above the ground, as shown in Figure 1.

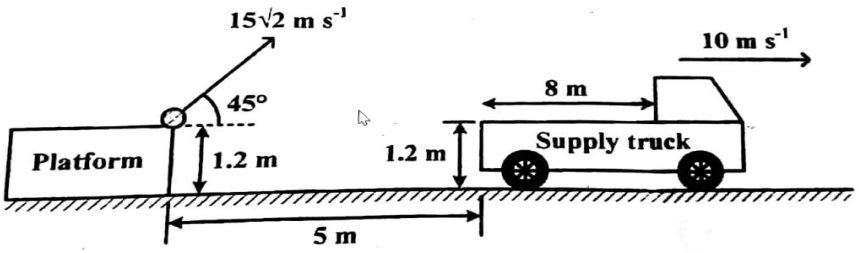


Figure 1

The emergency response agency will continue using the current loading procedure if the medical package lands inside the cargo bed of the moving truck. Otherwise, the loading procedure will be reviewed and adjusted before the next package is launched.

Hint

Take acceleration due to gravity, g = 9.8 m s^{-2}.

Task

Using appropriate principles of mechanics and mathematical calculations, advise the emergency response agency on:

(a) Whether the airport radar must be deployed to continue monitoring the aircraft.
(b) Whether the planned flight paths should be approved or one aircraft should be assigned an alternative flight route.
(c) Whether the current loading procedure should be maintained or reviewed and adjusted before subsequent packages are launched.

MATIGO

ITEM 15

A businessman wants to install a regular pentagon solar panel structure. He consults an engineer who finds out that the regular pentagon solar panel structure has a mass of 10 kg and a length of 2 m, with forces of $6\,N$, $13\,N$, $5\,N$, $2\,N$, $and\,3\,N$ acting along the sides AB, BC, CD, DE, $and\ AE$ respectively. The directions of the forces are in the order of the letters. The engineer took AB as the reference horizontal axis. The engineer advised that there was need to know the resultant force and that it should not exceed 18 N for the cost of the solar panel ($ABCDE$) to be affordable. It was also important to know the direction of the resultant force.

The engineer informed the businessman that the regular pentagon solar panel was installed at the top of the roof at a distance 4 m from the edge of the roof that is inclined at 30° to the horizontal. There was need to know the velocity with which it reaches the edge in case it is detached from the roof and slides down a rough surface with a coefficient of friction of $\frac{1}{4}$. He advised that the velocity should not go beyond 10 $m\,s^{-1}$, otherwise there would be loss of energy.

The engineer noted that the regular pentagon solar panel was projected below the horizontal at an angle of 30° below the horizontal with a velocity above. The height above the ground to the edge was measured as 4 m by the engineer. There was need to know the horizontal distance from the height to where the rectangular solar panel hits the ground. He preferred that it does not go beyond 3 m since there are usually children playing nearby. There is need to know the safety of these children.

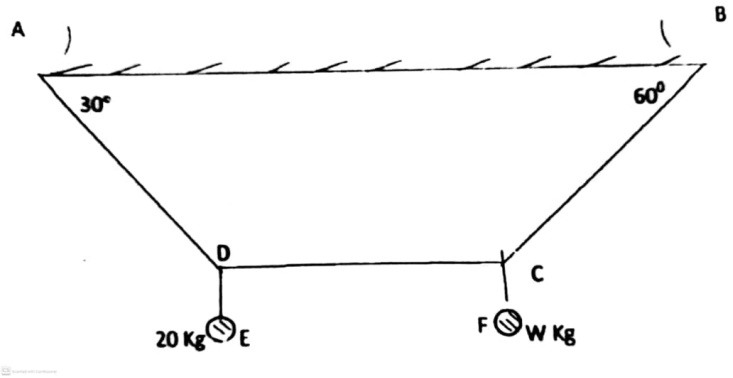
Task

As a mathematics student, analyze and justify the regular pentagon solar panel problems to the businessman.

ASSHU- ANKOLE

ITEM 16

A hotel management has a lighting system that is being investigated. It consults its engineer, who informs them that the lighting system is attached to three strings AD, DC, and BC. The strings AD and BC are fixed to points A and B. Bulbs of mass 20 kg and W kg are attached at points D and C respectively and rest in equilibrium position as shown below.



The engineer advised the hotel management that there was need to determine the tensions in the strings AD, DC, BC, and the weight. If W exceeds 15 kg, there is likelihood that the string will break.

The engineer observed that the string broke and the particle at F slid from rest down a rough inclined plane that is at an angle of 30° to the horizontal, which had a coefficient of friction of ⅕. It slid for a distance of 6 m. He advised the hotel management that there was need to know the velocity on reaching the lowest point of the plane, and if it is less than 15 m s^{-1}, there would be no destruction of the bulb.

The engineer observed that on reaching the ground level, it slid on a rough horizontal plane which had a coefficient of friction of $\frac{1}{4}$, with a force of 130 N, and moved until it came to rest. There is a wall at a distance of 10 m, and the engineer advised that there was need to know the safety of the wall.

TASK

As a mathematics student, help the hotel management to interpret, use a suitable methodology, and give justification on the findings of the engineer.

ASSHU- ANKOLE

ITEM 17

A trader in Kasese who runs a dairy farm finds challenges in delivering milk uphill to where his dairy shop is located. Since her farm is situated down at the foothill of Rwenzori Mountain, he was advised to buy a container and fasten 4 ropes on it to enable 4 men to lift it to the dairy uphill. The ropes were fastened in such a way that each was in the direction $N\ 60°\ E,\ 320°,\ 520°\ E,\ and\ 210°$. The men could provide forces in the ropes equivalent to $30\ N,\ 20\ N,\ 70\ N,\ and\ 60\ N$ respectively.

In the long run, this seemed costly and time-wasting. He was advised instead to use a pulley placed at the top of the hill driven by an engine. This made him buy a larger tank of mass 20 kg to carry milk weighing 100 kg and a cable that would pass over a smooth fixed pulley connecting the tank to another mass of 10 kg that could act as a counterweight. The

tank containing milk moves up a slope of 3 in 5 that is rough, with a coefficient of friction 0.4. The counterweight hangs freely vertically downwards on the other side of the hill to control downward motion for refill.

The pulley should be strong enough to withstand the force imposed on it by the cable for the combined masses uphill, and he also wishes that when the engine is switched off, the empty tank should slide down with an acceleration of $1.896\ m\ s^{-2}$ under the influence of a force P_{n} acting downwards parallel to the plane.

Task

As a mechanics student, if consulted, use clear calculations to come up with proper advice on:

(a) i. A single force and its direction that is equivalent to the system of forces imposed on the container by the four men.
ii. Whether buying a pulley that could withstand a total net force of 180 N for the combined mass uphill would work well for him.
(b) The magnitude of force P required in achieving the required acceleration. How safe can this system be kept working efficiently? *(Give scientific ways.)*

COUHEIA

ITEM 18

A game ranger at Murchison Falls National Park spots a buffalo running along a straight line towards River Nile. The ranger observes it at post A. At the moment of observation ($t = 0$), the buffalo is at point B,120 m east of post A and is running due north at a constant speed of 20 m s^{-1} relative to the ground.

Simultaneously, a rescue drone is launched from post A to move as close as possible to the buffalo's path. The drone flies at a constant speed of 10 m s^{-1} relative to the ground, maintaining a fixed direction.

At the instant when the drone is $60\sqrt{3}\ m$ horizontally away from the buffalo, it projects a tranquilizer dart horizontally towards the buffalo. The dart is released from a height 80 m above the ground with an initial speed $u\ m\ s^{-1}$. To avoid injuring the buffalo, the speed with which the dart hits the buffalo should not exceed 47 m s^{-1}.

When the buffalo was hit, it got shocked and jumped upwards a distance of 0.45 m high. The game ranger wishes to get analysis of the above information mathematically for better reporting.

Tasks

As a dynamics student, come up with necessary calculations to help the ranger establish:

(a) The closest distance reached by the drone and the time it took to be achieved.
(b) If the buffalo was injured or not.

(c) The initial velocity with which the buffalo jumped.

COUHEIA

ITEM 19

Peter is a very concerned parent. One day, he bought two toy cars , one of mass 1 kg and the other of mass 4 kg. To his surprise, he found his 5-year-old boy trying to combine the two toys, but the strings kept breaking and sometimes the heavier toy would fall on his feet. This always made the little boy cry a lot while playing. Peter decided to get him a rough plane and tried to arrange the setup as follows:

For the safety of the child, he placed the 1 kg mass toy on a 4 m long rough plane which has a coefficient of friction 0.25. A smooth fixed pulley was placed at the edge of the plane. A light inextensible string was connected to the 1 kg mass and allowed to pass over the pulley to a 4 kg mass which hangs freely at 2.5 m above the horizontal ground. Since he wanted the boy to enjoy the movement of the toys, he tilted the plane such that the angle of inclination θ with the horizontal is $sin^{-1}\left(\frac{3}{5}\right)$.

The system is released from rest. In order for Peter's son not to cry again, the following conditions must be met:

Condition 1: Acceleration of the system must be less than 7 $m\,s^{-2}$

Condition 2: The tension in the string must not exceed 15 N

Condition 3: The maximum kinetic energy with which the 4 kg mass hits the ground must be less than 70 J

Task

As a mathematics student, determine whether the setup Peter made will stop his son from crying and justify your responses.

UTES

ITEM20

Pirates always smuggle people's goods in oceans and seas. Due to that, shipmen and fishermen always try to avoid the paths which they like.

One businessman always imports his products from India to Mombasa via the Indian Ocean. He uses a global positioning system (GPS) to track the path, speed, and position of the ships carrying his goods. One day, when he was checking the progress of his goods, he observed that at 11: 30 $a.m.$ a pirate ship is at a place with position vectors $(-6i + 12j)\ km$ and is moving with velocity vector $(16i - 4j)\ km/h$. At 12: 00 $noon$, the ship carrying his goods is at a place with position vectors $(12i - 15j)$ km and is moving with velocity vector $(8i + 16j)\ km/h$.

Both ships' velocities are constant, and the businessman is worried that the pirate might destroy his goods if they meet. However, he sent a rescue team which must reach their ship before 1 p.m.

When the rescue team saw the pirate ship, one of the soldiers in the rescue ship fired a bullet with a velocity of 320 m/s at an angle of 30° to the horizontal. But the bullet must pass over a vertical cliff in order to hit the right target, and the bullet must be shot when the ships are less than 10,000 m from each other.

Task

As a student of mathematics: Determine whether the businessman's goods will reach Mombasa safely or not. Justify your response.

UTES

ITEM 21

A mechanical engineer visited a vocational training workshop to test three mechanics demonstrations about force systems, friction, and pulleys.

On a horizontal table, a particle was acted on by forces of magnitudes $30\ N, 50\ N, 60\ N, 100\ N, and\ 20\ N$ at the same time, acting in the directions $060°, 090°, 120°, 210°, and\ 330°$ respectively. The engineer needed to know the net effect of the forces acting on the particle and its direction.

In another section, a 4 kg block was pulled across a rough horizontal plane using a string inclined at an angle of 20° above the horizontal. The tension in the string was 25 N, and the block accelerated at $4\ m\ s^{-2}$. The engineer noticed that the nature of the horizontal surface was affecting the motion of the block and wanted to determine the coefficient of friction between the block and the surface.

In the pulley section, a light inextensible string attached to the ceiling passed under a smooth movable pulley of mass 3 kg, then over a smooth light pulley fixed at the top of an inclined plane, and was finally attached to a body of mass m kg resting on a rough inclined plane set at 30° to the horizontal. The coefficient of friction between the body and the plane was 0.25, and when the system was released from rest, the m kg mass covered a distance of 2 m in 4 seconds along the plane while the movable pulley moved vertically. The engineer wanted to decide whether the unknown mass was light enough to be used in the demonstration if the maximum allowed mass on the inclined plane was supposed to be 8 kg.

Task

As a mathematician, use the above information to determine the net effect of the forces acting on the particle and its direction, the frictional force and coefficient of friction and decide whether the body on the inclined plane is safe for the demonstration.

MEC

ITEM 22

A medical emergency has broken out in Northern Uganda, and a humanitarian organization based in Gulu is coordinating the urgent delivery of vaccines and medical supplies to a remote health centre in Kitgum.

At the health centre, a nurse is stationed at a window on the upper floor of the hospital building waiting to receive a small emergency package. A health worker stands on the ground 10 m from the hospital building and throws the package from a height of 1.0 m above the ground with a speed of 11.5 m s^{-1} at an angle of 60° above the horizontal. The lower edge of the window is 3.2 m above the ground while the upper edge of the window is 4.5 m above the ground. The nurse needs to know whether the package will pass through the window or strike the wall of the hospital building.

At the same time, a medical aeroplane takes off from airstrip A in Gulu and must fly directly to airstrip B in Kitgum, which is 500 km due south of airstrip A. A steady wind of 100 km/h blows from the east while the aeroplane can fly at 400 km/h in still air. The pilot does not know the right course to set so that the aeroplane moves directly to airstrip B.

At exactly 1500 hours, while the aeroplane is en route, air traffic controllers detect a rescue helicopter 80 km on a bearing of 290° from the aeroplane. The helicopter immediately flies on a bearing of N 30° E at a speed of 15 km/h. At the same instant, the aeroplane travels at 12 km/h on a bearing of N 300° W. The control tower operator must report to her supervisor on the possible closest distance between the two aircraft and the exact time when this occurs.

Task

As a mathematician, help the nurse to know if the package will pass through the window or strike the wall of the hospital building and its speed when it reaches the building. Set the right course the pilot needs to fly the plane directly to airstrip B, and determine the facts the control tower operator needs to report to her supervisor about the motion of the two aircraft.

MEC

ITEM 23

At a stone quarry, engineers are transporting stones using a lorry pulling a trailer, while at the same time lifting stones to an upper platform using a pulley system. They must also ensure that the forces acting on the loading structure are balanced for safety.

A lorry of mass 2000 kg is pulling a trailer of mass 1500 kg along a horizontal road. The driving force produced by the lorry engine is 5000 N, and the resistances to motion of the lorry and trailer are 600 N and 400 N respectively. The engineers want to determine the acceleration of the system and the tension in coupling between the lorry and the trailer. If the tension exceeds 2500 N, the coupling may fail.

At the loading point, a stone of mass 40 kg lies on a rough inclined plane inclined at an angle $sin^{-1}\left(\frac{1}{2}\right)$. It is connected by a light inextensible string passing over a smooth fixed pulley to another mass of 60 kg hanging vertically. The coefficient of friction between the stone and the plane is 0.3. If the pulley system moves too fast with acceleration greater than 2 m s^{-2}, the lifting mechanism becomes unsafe.

The pulley is attached to a loading frame that is being stabilized by three cables exerting forces of 30 N acting due East, 40 N acting due North, and 50 N acting at an angle of 60° West. If the resultant force exceeds 70 N, an additional support must be installed.

Task

Help the engineers determine whether the coupling will fail or not, whether the lifting mechanism is safe or not, and whether the additional support is needed or not.

TAEB

ITEM 24

At exactly 11:00 am, an emergency response coordinator was monitoring two rescue units during a disaster response exercise. The emergency control centre was taken as the origin.

A taxi was carrying medical supplies while a rescue truck was transporting emergency equipment. Their initial positions and constant velocities at 11:00 am were recorded as follows:

Unit	Position vector	Velocity
Taxi	$(5i + 2j)\ km$	$(15i + 10j)\ km/h$
Rescue truck	$(7i + 7j)\ km$	$(9i - 5j)\ km/h$

The coordinator wanted to determine whether the two rescue units would arrive at the same location if they continued moving with their recorded velocities. If they would, determine the time when they would meet and the total distance travelled by the rescue truck by that time.

Later, the taxi developed a mechanical fault and had to be transported to the maintenance centre. A recovery truck of mass 1800 kg was used to tow the damaged taxi, which was of mass 1200 kg, using a tow bar along a rough horizontal road.

The recovery truck applied a driving force of 6000 N. The damaged taxi experienced friction with a coefficient of 0.2, while the recovery truck experienced a resistive force of 800 N.

For the towing operation to be considered safe, the acceleration must be less than 2 m s^{-2} and the tension in the rope must not exceed 3500 N. Otherwise, a heavier recovery vehicle would be required.

Task

As a student of mathematics, help the coordinator to make informed decisions.

TAEB

ITEM 25

A school located on a hilly compound is preparing for a safety and emergency response exercise. During the exercise, the school maintenance team wants to deliver a first-aid package to learners standing near a raised classroom block. However, the area around the classroom is narrow, and the package must be lowered carefully to avoid hitting the wall, injuring learners, or missing the intended delivery point.

First, the package of mass 8 kg is temporarily suspended from a fixed beam at point A using a light inextensible string of length 50 cm. To keep the package clear of the classroom wall before it is released, a horizontal force P is applied so that the package remains at 30 cm from the vertical line through A. The maintenance team wants to know whether the arrangement is safe, but they are not sure whether the string can withstand the pull. The string is considered safe only if its tension does not exceed 100 N, and the horizontal pull is considered manageable only if $P < 70\ N$.

At the same time, two emergency response groups are approaching the classroom block from different paths within the school compound. The first group moves with a velocity of $8\ km\ h^{-1}$ in the direction $N\ 80°\ W$, while the second group moves with a velocity of $5\ km\ h^{-1}$ in the direction $S\ 25°\ W$. The school coordinator wants to guide them towards a common meeting point, but is not sure about their combined direction of movement. For the movement to be considered well controlled, the resultant speed should not exceed $10\ km\ h^{-1}$, and the direction of the resultant velocity should be clearly determined.

Finally, instead of carrying the package through the crowded corridor, the team plans to project a small emergency kit from level ground to a balcony where another teacher is waiting to receive it. The balcony is 6 m above the ground and 18 m horizontally from the point of projection. The kit is projected with an initial speed of $20\ m\ s^{-1}$ at an angle of 40° to the horizontal. The team wants to know whether this method of delivery is suitable, but they are not sure whether the kit will pass close enough to the balcony position at a height between 5.5 m and 6.5 m, and if the time taken to reach the balcony is less than 2 seconds.

Task

As a student of mathematics, help the emergency team to analyze the situation and make suitable safety decisions.

WENSSEC

ITEM 26

At 12 noon, two vehicles A and B are moving on a flat road near a busy junction. Their positions and velocities are being monitored from a traffic post, which is taken as the origin. The position vectors and velocity vectors of the vehicles are given by:

$$r_A = \begin{pmatrix} 5 \\ 2 \end{pmatrix}\ km, V_A = \begin{pmatrix} 15 \\ 10 \end{pmatrix}\ km\,h^{-1}$$

$$and$$

$$r_B = \begin{pmatrix} 7 \\ 7 \end{pmatrix}\ km, V_B = \begin{pmatrix} 9 \\ -5 \end{pmatrix}\ km\,h^{-1}$$

The traffic officer wants to know whether the two vehicles will collide if they maintain their present velocities. The situation is considered dangerous only if the vehicles meet at the same position within the next 30 minutes. If a collision occurs, the officer also wants to know the exact time after 12 noon and the position vector of the collision point, so that emergency support can be directed to the correct location.

After the collision, vehicle B is badly damaged and cannot move on its own. A traffic police vehicle is used to tow it along a rough horizontal road to a safe parking area. The damaged vehicle has a mass of 1200 kg, while the towing police vehicle has a mass of 1800 kg. The two vehicles are connected by a light inextensible tow rope. The police vehicle applies a driving force of 6000 N. The coefficient of friction between the damaged vehicle and the road is 0.20, while the resistance on the police vehicle is 800 N.

The towing process will be considered safe only if the acceleration of the two vehicles is less than 2 m s^{-2} and the tension in the tow rope does not exceed 3500 N. The traffic officer is not sure whether the police vehicle can tow the damaged vehicle safely without causing further damage to the tow rope or losing control of the motion.

Task

As a student of mathematics, help the traffic officer to analyze the collision risks, locate the possible collision point, and decide whether the towing process is safe.

WENSSEC

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